Govardhana Kondapaturi

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SUMMARY

Physical Design Engineer with 3+ years across AMD and Astera Labs, currently closing timing on 7 nm high-speed connectivity silicon. Drive the implementation flow from floorplan through CTS, routing, MCMM signoff, and ECO in Cadence Innovus and Synopsys PrimeTime, with Python/TCL and LLM-augmented automation. Led a 180K-gate three-chiplet SoC with 12 asynchronous clock domains from RTL to GDSII at 513 MHz (IEEE LASCAS 2027 submission) and submitted an all-digital PLL to the SkyWater SKY130 shuttle on September 7, 2026. M.S. ECE, California State University, Fresno.

TECHNICAL SKILLS

- **Physical Design & Implementation:** Cadence Innovus, floorplanning, macro placement, power/clock distribution, CTS, routing, congestion analysis, timing closure, ECOs, multi-voltage design (UPF, clock gating, multi-Vt, isolation, level shifters)
- **Static Timing & Physical Verification:** Synopsys PrimeTime, MCMM setup/hold analysis, SDC constraints, signoff STA, RedHawk-SC IR-drop and EM, Calibre DRC/LVS/PEX/ANTENNA/ERC signoff, gate-level simulation with SDF annotation
- **RTL, Simulation, Synthesis & Formal:** Verilog, SystemVerilog, Cadence Xcelium simulation, Synopsys Design Compiler, DFT-aware synthesis, SpyGlass CDC/RDC and lint, JasperGold, Conformal and Formality equivalence checking
- **Programming & Automation:** Python, TCL, Perl, C/C++, STA report parsing, ECO/regression automation, LLM-augmented tooling

PROFESSIONAL EXPERIENCE

Physical Design Engineer | Astera Labs
San Jose, CA, USA | Aug 2025 – Present

- Implement 7 nm ASIC physical design in Cadence Innovus: floorplanning, power planning, placement, CTS, and routing.
- Run MCMM timing analysis in Synopsys PrimeTime and resolve setup and hold violations across PVT corners.
- Analyze placement congestion and routing issues affecting critical timing paths in high-speed connectivity designs.
- Apply cell-sizing, buffering, and ECO changes to close timing violations and improve timing margins.
- Verify clock-tree skew, insertion delay, and transition violations during timing closure.
- Compare pre-route and post-route timing reports to isolate delays caused by parasitic effects.
- Review physical signoff results, including DRC, LVS, antenna checks, IR drop, and electromigration.
- Use TCL and Python scripts to analyze timing reports and automate recurring implementation checks.

Silicon Design Engineer | AMD
Bengaluru, Karnataka, India | Jun 2022 – Jul 2024

- Analyzed ASIC floorplans and placement results to identify congestion hotspots and implementation changes.
- Investigated critical timing paths in Synopsys PrimeTime and worked through setup and hold violations across PVT corners.
- Evaluated clock-tree reports for skew, insertion delay, and transition violations affecting timing closure.
- Examined post-route timing reports and extracted parasitics to determine their impact on signal delays.
- Troubleshot physical verification violations involving DRC, LVS, and antenna checks through to clean signoff.
- Developed TCL scripts to extract timing metrics, organize implementation reports, and reduce repetitive analysis.
- Coordinated timing, placement, and verification findings to track open physical implementation issues to closure.

ACADEMIC PROJECTS

All-Digital PLL Tapeout on SkyWater SKY130 (TinyTapeout SKY26c Shuttle) **Sole Designer | Aug – Sept 2026**

- Designed an all-digital PLL (TDC, PI loop filter, ring-oscillator DCO, programmable divider) in synthesizable Verilog.
- Hardened in OpenLane/OpenROAD on SKY130: 40 MHz, sub-10% IR drop, clean DRC/LVS/ANTENNA via Magic/KLayout.
- Submitted GDSII to the ChipFoundry MPW shuttle on September 7, 2026; silicon expected May 2027.

Multi-Chiplet SoC with DVFS, CDC, and Hierarchical Power Management **Team Lead | Jan – May 2025**

- Led RTL-to-GDSII of a 180K-gate SoC (app processor, NPU, shared L2 over NoC) with 12 async clock domains in Innovus.
- Closed 513 MHz with MCMM PrimeTime signoff; built UPF multi-voltage domains with DVFS, isolation, level shifters.
- Root-caused a 30 ps hold on die-to-die NPU-to-L2 AXI at 0.55 V; fixed by local skew rebalance, no setup regression.
- Reached 100% CDC/RDC closure in SpyGlass and JasperGold with clean Conformal equivalence against synthesized netlist.

EDUCATION

Master of Science in Electrical and Computer Engineering **Fresno, USA**
California State University | GPA: 3.3/4.0

Bachelor of Technology in Electronics and Communication Engineering **Vijayawada, India**
Vasireddy Venkatadri Institute of Technology | GPA: 3.56/4

PUBLICATIONS

- G. Kondapaturi and A. Stillmaker, "A UPF-Driven Multi-Voltage Three-Chiplet SoC with DVFS, carried from RTL to GDSII," submitted to IEEE LASCAS 2027.

CERTIFICATIONS

- **Synopsys:** PrimeTime, Fusion Compiler Design Implementation.
- **Cadence:** RTL-to-GDSII Flow v7.0 (Advanced), SystemVerilog for Design and Verification, Static Timing Analysis v3.0.